ROMIL SHAH

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SUMMARY

Lead Research Scientist with expertise in building frontier LLMs from research to production at Meta, leading Agentic Memory and Continual learning capabilities. Built RL training environments, reward models, and data flywheel systems powering post-training across Llama and Muse model series to enable continual learning of Meta AI agents.

EXPERIENCE

Lead Research Scientist, Meta Superintelligence Labs — Jan 2023 – Present

Post-Training Lead: Agent Memory, Continual Learning, User Modeling

Lead post-training research and production deployment powering the Personal SuperIntelligence goal for Meta AI. Specifically I drive the agentic search, memory, and continual learning pods for Meta's frontier LLMs.

- **Post Training for Agentic memory & tool use.** Architected and shipped agentic assistants with persistent identity and epistemology. Designed retrieval, context compression, and trigger-classification gating for long-horizon tasks. Built RL environments and reward models for tool-use and response quality.
- **Data flywheel & Continual Learning** Built end-to-end data flywheel from production traffic → preference data → continual model updates. Primary feedback channel powering ongoing post-training cycles.. Trained on RLHF, RLAIF, and online user-feedback signals — pairwise preference (side-by-side) and post-task surveys — shaping helpfulness, safety, and behavior across diverse axes.
- **Agentic web search post-training.** Drove post-training for Meta AI's agentic web search: query formulation, retrieval relevance, and grounding of generated responses to retrieved evidence. Reward shaping targeted hallucination control and faithfulness
- **Evaluation frameworks.** Built automated and human eval for tool-use accuracy, retrieval grounding, memory relevance, persona stability, and personalization quality. Continuous quality monitoring on live traffic.
- **Long-term user modeling.** Built a long-horizon user-profile foundation model used to condition personalization across Meta AI surfaces; enables stable, drift-aware personalization beyond a single session.
- **Distillation.** Compressed frontier-model capabilities into latency- and memory-constrained serving targets while preserving alignment and behavioral properties; productionized for long-tail deployments.

Staff Research Scientist, Meta — Jan 2019 – Jan 2023

Core Modeling Lead, Meta Reels & Video (2B+ users)

Generative Recs, RL based alignment and User Understanding

- **RL based Slate Ranking :** Pioneered RL-based recommendation for interest exploration and long-term value optimization at Meta scale. Published methodology at ACM WebConf '23 (PIE).
- **Generative Recommendations.** Introduced generative recommender paradigm for video engagement, driving >10% engagement gains across Facebook Video.
- **User Understanding.** Pioneered foundational user model pretraining used across all Meta recommendation surfaces

Principal Engineer, Bloomreach Inc. — Jan 2013 – Oct 2018

- Architect of Search Quality, Personalization, and Analytics. Led transition from keyword to NLP semantic ranking with pairwise learning-to-rank; built behavior-derived product knowledge graph. Avg. 12% revenue lift across customers; grew team 2 → 12.
- Authored core search-attribution and A/B-testing infrastructure foundational to Bloomreach analytics today.

Senior Software Development Engineer, Microsoft — Oct 2010 – Dec 2012

Shipped Windows OS features: Natural Language System, Windows Search, and user-environment components.

Software Engineer, Cleartrail Technologies — Aug 2008 – Oct 2010

Built distributed real-time telecom monitoring and high-throughput call processing pipelines.

PATENTS & PUBLICATIONS

- PIE: Personalized Interest Exploration for Large-Scale Recommender Systems WWW'2023
- Query-dependent and content-class based ranking US 9830392 · Issued Nov 28, 2017
- Search Ranking US 20180060936 Filed Mar 1, 2018

SELECTED SKILLS

Post-training & alignment: RLHF, RLAIF, DPO, reward modeling, preference data pipelines, model distillation, safety tuning.

Agentic systems: tool-use training, agentic memory, retrieval, context compression, inference-time tooling.

Infrastructure: PyTorch, async distributed RL training, large-scale RL environments, automated + human eval frameworks, A/B testing.

Languages: Python.

EDUCATION

Faculty of Technology, Dharamsinh Desai University, India
Bachelor of Engineering, Computer Science